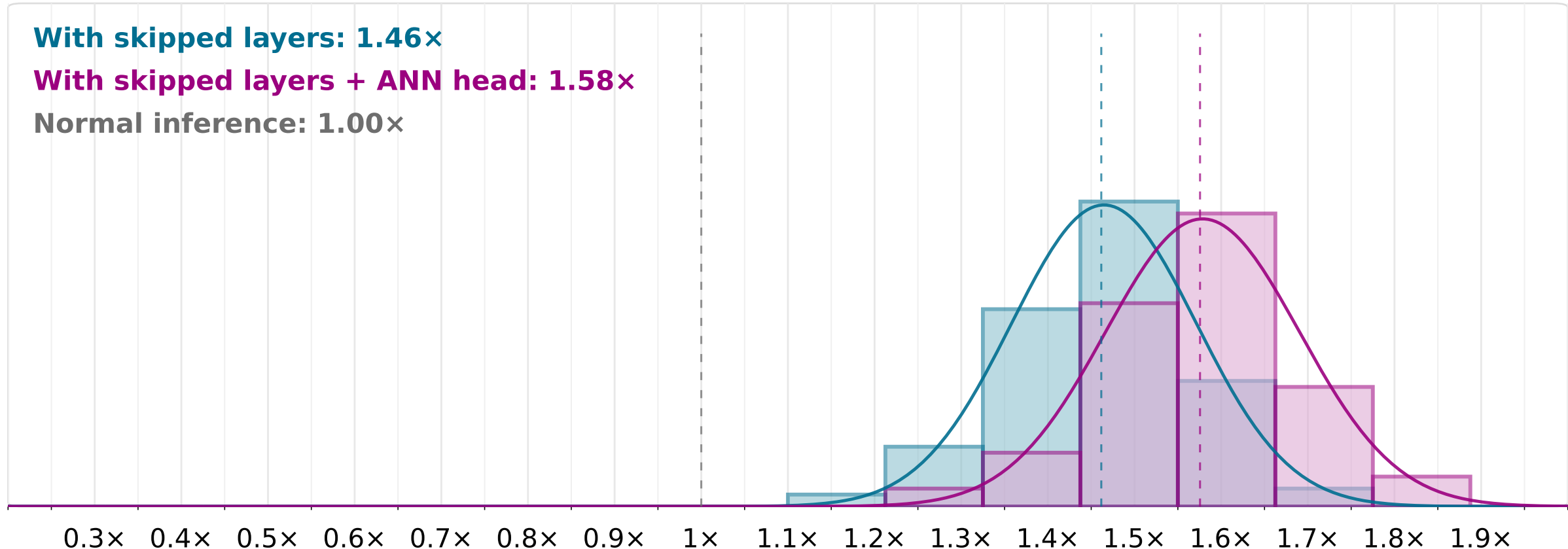


Self-speculation per-token speedup

Llama 3.1 8B Instruct



Setup

Prompt set = **concrete**
Gap = **(1, 1)**
Block size = **2**
Layers = **32**
LM vocab = **128,256**
LM params = **8.03B**
Head params = **525.3M**
Head portion = **6.54%**

Runtime

GPU = **NVIDIA L4**
Backend = **sdpa**
Model dtype = **bfloat16**
Bridge dtype = **bfloat16**
Speed internals = **no**
Profile prompts = **15**
Warmup prompts = **5**
Measured prompts = **120**

Results

Peak mem normal = **15.06 GiB**
Mean speedup (w/o ANNH) = **1.46×**
Acceptance rate (w/o ANNH) = **47.7%**
Exact match (w/o ANNH) = **43.3%**
Peak mem self (w/o ANNH) = **15.06 GiB**
Mean speedup (w ANNH) = **1.58×**
Acceptance rate (w ANNH) = **46.9%**
Exact match (w ANNH) = **43.3%**
Peak mem self (w ANNH) = **15.14 GiB**
Tokens gen = **s 23,452 / n 23,686**

Profile

Drafter split (w/o ANNH) = **B 49.1% / H 46.0% / O 4.8%**
Verifier/normal (w/o ANNH) = **1.05×**
Drafter/normal (w/o ANNH) = **14.4%**
Drafter split (w ANNH) = **B 82.0% / H 10.2% / O 7.7%**
Verifier/normal (w ANNH) = **1.05×**
Drafter/normal (w ANNH) = **8.5%**
ANNH accuracy = **98.4%**
ANNH head speedup = **7.47×**
ANNH index = **8016 clusters; top-k 100**